

UNIT 3: CELLS

3.5 – STEM CELLS & CANCER

Questions

- What is the purpose of a stem cell?
- What in the cell cycle makes cancer different than normal cells?

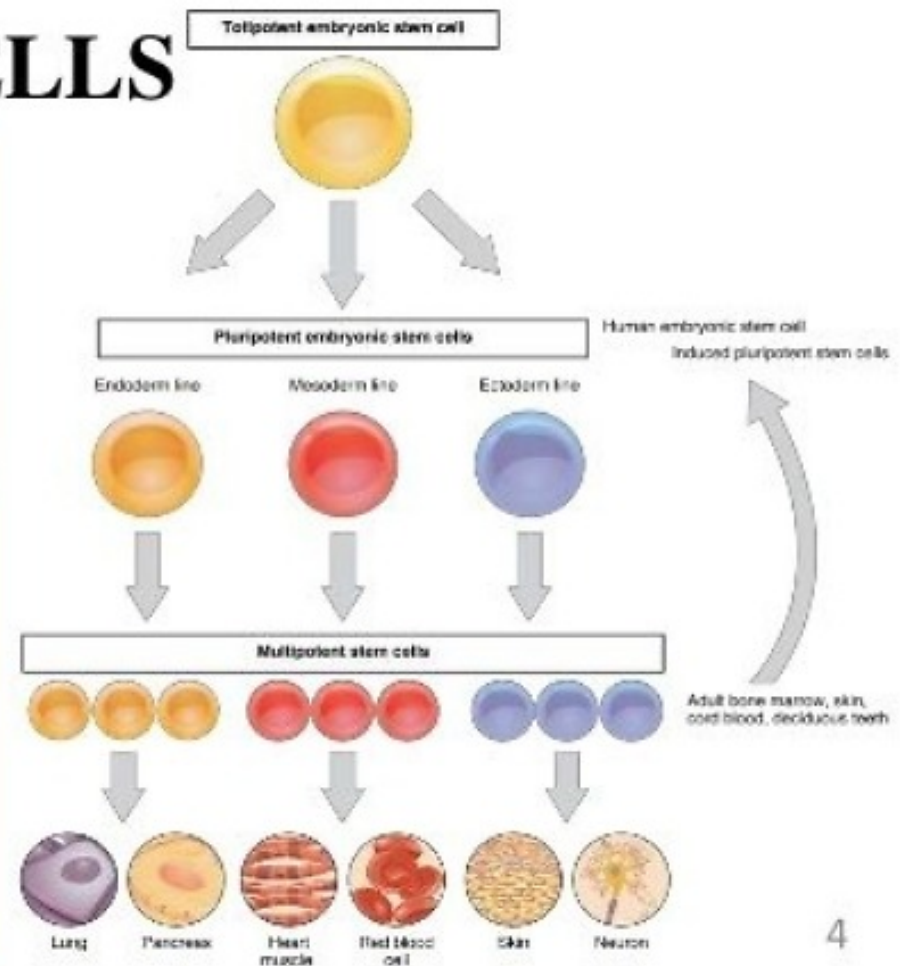
Stem Cells

- Most cells of the body in adulthood permanently arrested in G_0 stage
- If injury occurs, body must be able to repair the injury (cut, broken bone, etc)
- Some cells must retain ability to divide
- **Stem cells** – Cells that retain the ability to divide and differentiate
 - ▣ **Embryonic** – from blastocyst (can become any cells)
 - ▣ **Tissue Specific** – somatic
 - **Hematopoietic** - all blood cells
 - **Mesenchymal** – in bone marrow; tissue, bone, cartilage, fat repair
 - ▣ **Induced pluripotent** – lab engineered from somatic cells; act like embryonic, can become all cells

Stem Cells

TYPES OF STEM CELLS

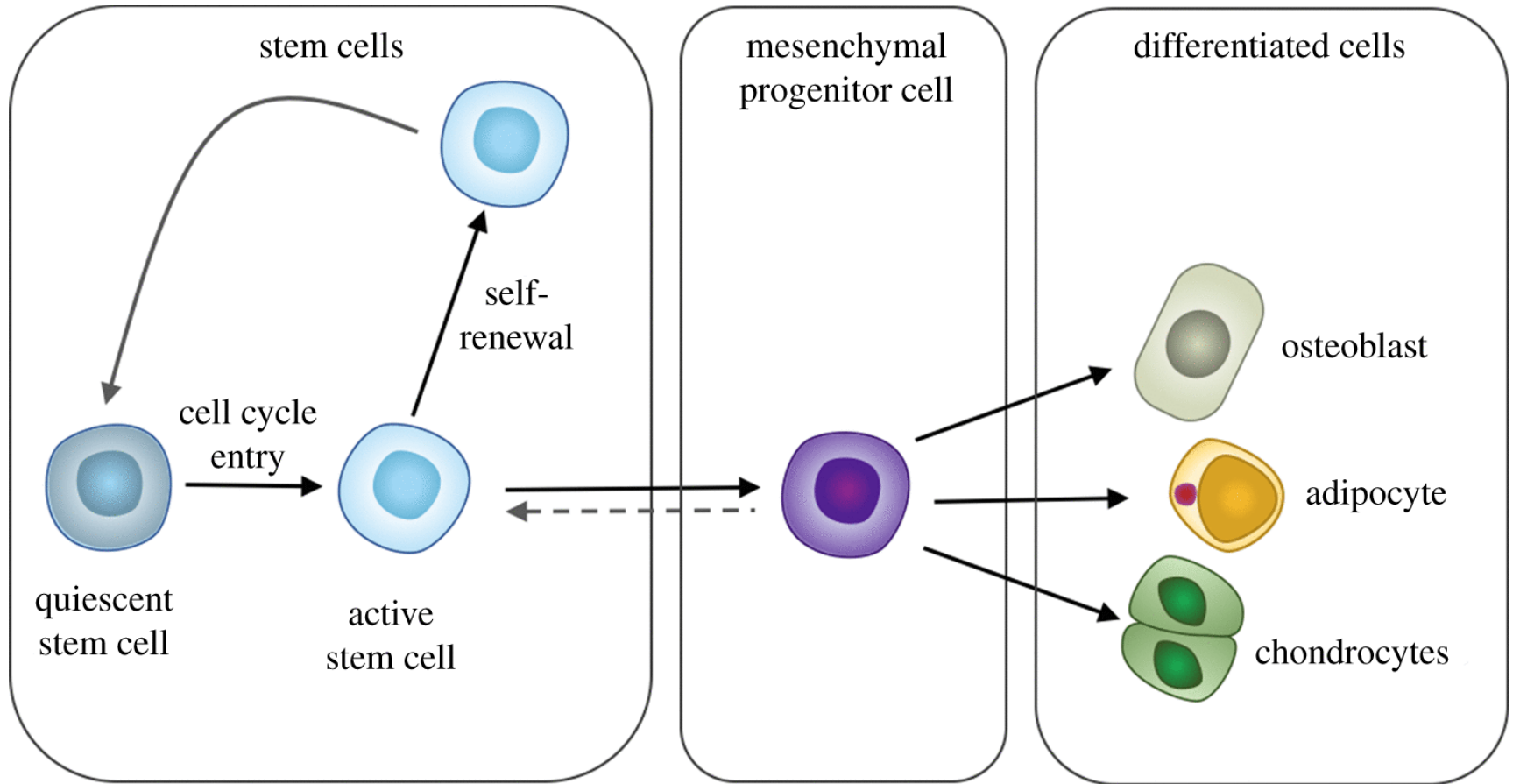
Type of stem cell	What it can be	Examples
Totipotent cells	Each cell can develop into a new individual	Cells of embryo of 1-3 days
Pluripotent cells	Each cell can form any cell type (over 200)	Cells of blastocyst 5-14 days
Multipotent cells	Cells differentiate and can form a number of tissue types.	Fetal tissue, cord blood, adult cells



Stem Cell Differentiation

- As stem cells begin to divide based on the signals they receive, they begin to **differentiate**, or become more specialized
- Method for producing cells that perform specific functions in the body, instead of all cells needing to do all functions
 - ▣ Cardiac cells vs nerve cells vs skin cells
- Stem cells can remain stem cells through replication. Once differentiation occurs, they terminally differentiate (cannot return to stem cells)

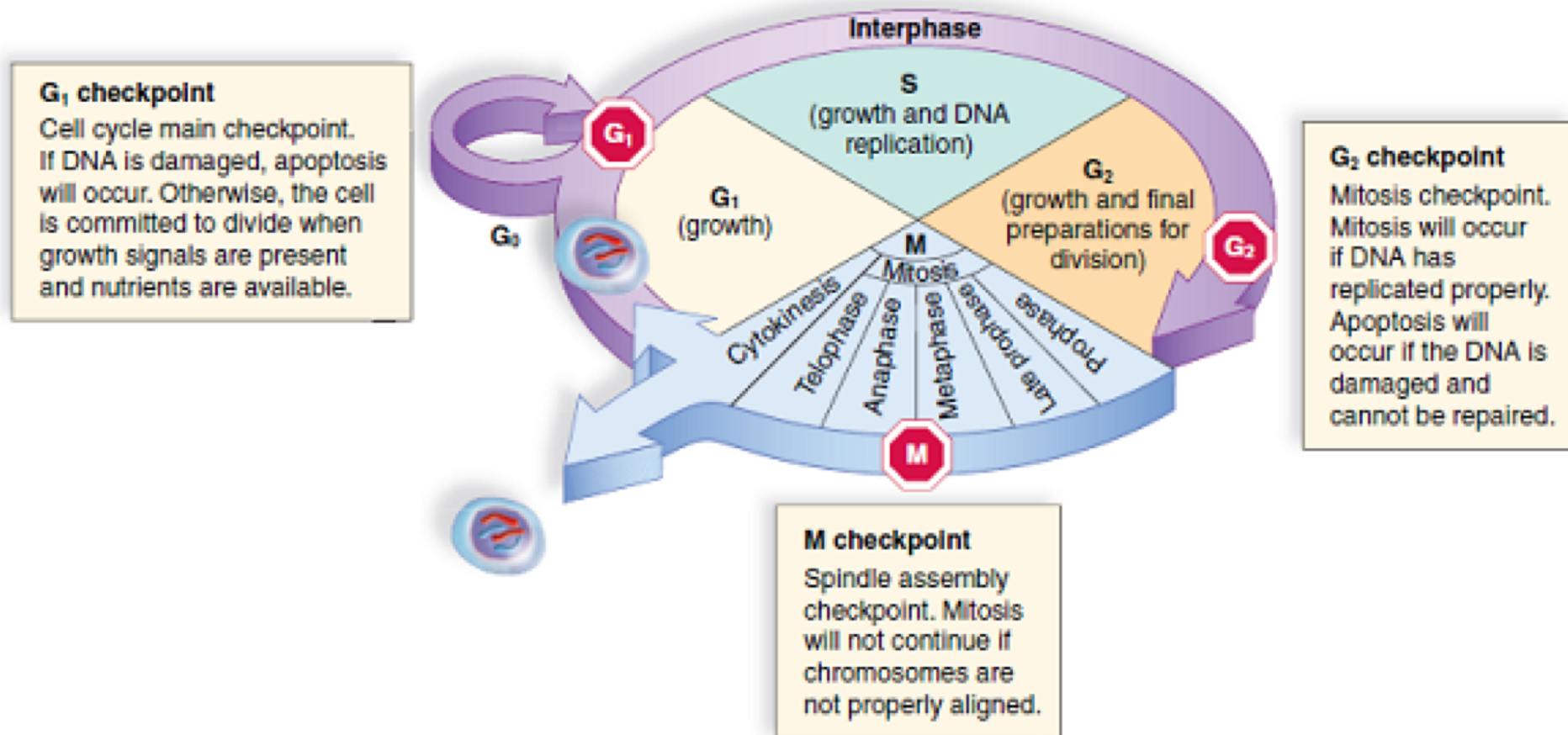
Stem Cell Differentiation



Control of the Cell Cycle

- A **signal** is an agent that influences the activities of a cell.
- Growth factors are a type of signaling proteins
 - ▣ Received at plasma membrane of a cell
 - ▣ Can push cell to continue through cell cycle, even if arrested in G_0
- Checkpoints
 - ▣ From previous lecture, cell cycle checkpoints at G_1 , G_2 , and M
- Cyclins
 - ▣ Internal signaling proteins
 - ▣ Increase and decrease as cell cycle continues
 - ▣ Specific cyclins must be present to move from G_1 to S phase, and from G_2 to M phase

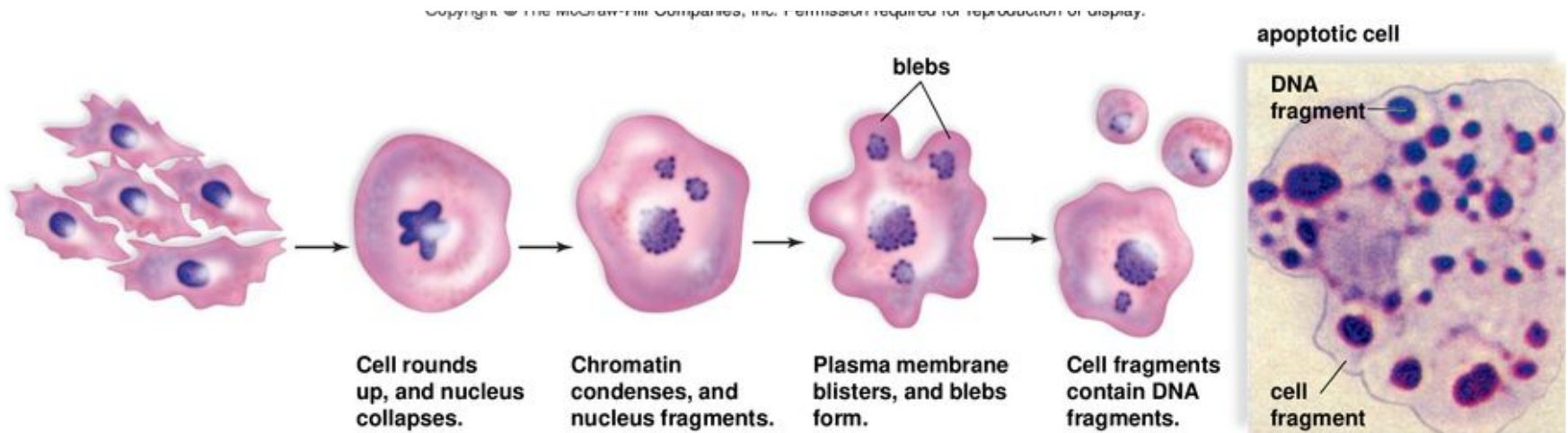
Control of the Cell Cycle



G₁ Checkpoint

- Primary checkpoint of the cell cycle
- **p53**
 - Protein 53kDa weight
 - Recognizes DNA damage and stops cell cycle at G₁
 - Tries to initiate DNA repair
 - If unsuccessful, or if p53 concentration builds, **apoptosis** is initiated
 - **Apoptosis** = Programmed cell death

Apoptosis



- Cell progresses through a series of events that lead to its destruction
- After going through steps above, white blood cells engulf cell fragments
- An enzyme class, **Caspases**, when not held in check by inhibitors, are signaled and lead to destruction

Apoptosis and Cell Division

- Cell division increases cell number and apoptosis decreases cell number (opposing processes)
 - Both normal parts of growth and development
 - Tadpole must lose tail to mature
 - Webbing between human digits
- Both constantly occurring throughout life
- Death through apoptosis prevents cells from continually growing, which can result in a tumor

Cancer

- Uncontrolled cell division
- Gradual process, resulting from the accumulation of mutations
- Once cells develop enough mutations, they develop distinct characteristics

1. **Cancer cells lack differentiation**

- ▣ Not specialized
- ▣ Possess characteristics of surrounding cells, but abnormal
- ▣ Can enter cell cycle indefinitely
 - Normal cells only capable of ~50 divisions

Cancer

2. **Cancer cells have abnormal nuclei**

- ▣ Nuclei enlarged, and sometimes have incorrect chromosome number, or duplicated portions of chromosomes
 - Results in gene amplification, or gene deletions

3. **Cancer cells do not undergo apoptosis**

- ▣ Mutations acquired may stop the checkpoint functions that would cause cells with damage to stop replicating and be destroyed

Cancer

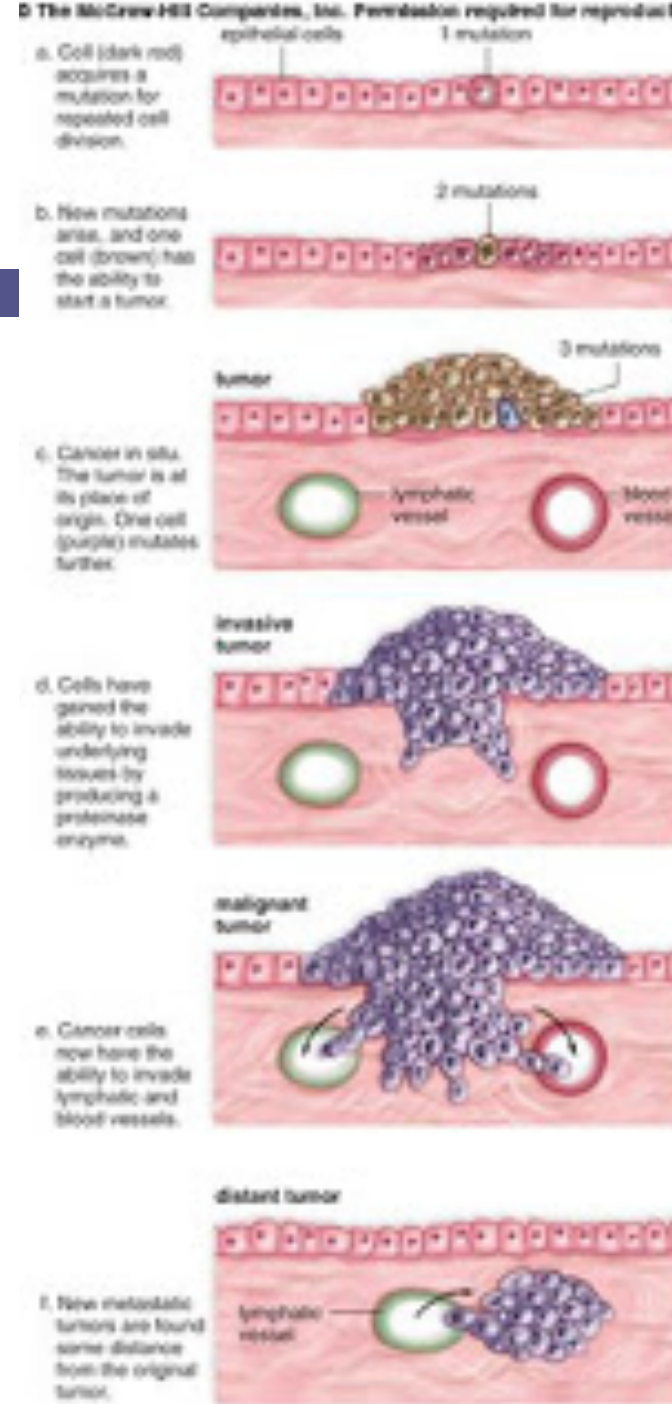
4. Cancer cells form tumors

- ▣ Normal cells stop dividing when they come in contact with other cells – **Contact inhibition**
- ▣ Cancer cells have lost this restraint

5. Cancer cells undergo metastasis and angiogenesis

- ▣ Invade underlying layers and can enter blood supply and spread to other areas of the body – **metastasis**
 - ▣ Acquired mutations may allow cells to signal the creation of new blood vessels to supply it nutrients to grow – **angiogenesis**
-
- ▣ Cancer treatments aim to attack some of these attributes

Cancer



Causes of Cancer

- **Proto-oncogenes**

- Proteins that promote cell cycle and prevent apoptosis
 - Cell Cycle Gas pedal

- **Tumor Suppressor Genes**

- Proteins that inhibit cell cycle and promote apoptosis
 - Cell Cycle Brake Pedal

- Mutations accumulated can affect the genes that code for these proteins

Causes of Cancer

- Mutation may cause over stimulation of oncogene to produce proteins, or gene or chromosome duplication can result in more copies of the gene
 - ▣ *ras* gene family most commonly affected
 - BRCA1 in breast cancer
- Mutations or chromosome deletions can result in abnormal or missing tumor suppressors
 - ▣ 1/2 of all human cancers have broken or missing p53 gene
- Many more examples you may find in future courses

Cell Cycle Regulation and Cancer

